

structure. Hence the structure must be known to find the paths that the waves took. To illustrate this, consider the travel time between two points. If the velocity were constant, the ray path would be a straight line, and the velocity could be found by dividing the distance by the travel time. If, instead, an interface separates media with different velocities, the ray path would consist of two line segments, depending on the velocities, and the travel time would be the sum of the time spent along each segment. For a more complicated velocity distribution, the ray path would also be more complicated.

This problem can be posed mathematically by writing the travel time between the source (s) and receiver (r) as the integral of $1/\text{velocity}$, or slowness, along the ray path

$$T(s, r) = \int_s^r \frac{1}{v(x)} dx. \quad (1)$$

In simple cases, where the ray path is a set of segments with constant velocity, the integral is just a sum over the time in each segment. Thus the travel time gives an integral constraint on the velocity distribution between the source and the receiver, but does not indicate which of the many paths satisfying the constraint the ray followed. As a result, an individual measurement is inadequate to show the distribution of velocities. Fortunately, as we shall see, a set of travel times between different sources and receivers provides much more information. In addition, useful information is derived from the amplitudes and waveforms of seismic waves.

This example illustrates an interesting feature of determining velocity structure from travel times. If the velocity structure is known, the forward problem of finding the travel times and amplitudes is straightforward. However, the inverse problem of using the travel times and amplitudes measured at the surface to find the velocity structure at depth is more difficult, and various methods are used. For example, in addition to using travel times directly, we have seen that velocity structure is studied using the dispersion of surface waves (Section 2.8) and the eigenfrequencies of normal modes (Section 2.9), quantities that correspond to travel times.

In this chapter, we follow the approach discussed in Section 1.1.2 of treating the earth with a series of progressively more complex and, hopefully, more accurate models. We begin with the homogeneous, isotropic, elastic, layered halfspace used in Chapter 2 to derive seismic wave propagation. This approximation of uniform flat layers is often used in crust and upper mantle studies, where the distance between source and receiver is less than a few hundred kilometers. We then consider larger source–receiver distances, for which spherical geometry is required, and then the anisotropic and anelastic behavior of the earth. Throughout these discussions, we will see that although velocity varies primarily with depth, there are important lateral variations, or heterogeneities. Finally, we consider the implications of the observed heterogeneous, anisotropic, and anelastic

velocity structure for the composition of the earth. Later, in Chapter 7, we discuss further how seismic data can be used to study laterally variable velocity structure.

3.2 Refraction seismology

3.2.1 Flat layer method

The simplest approach to the inverse problem of determining velocity at depth from travel times treats the earth as flat layers of uniform-velocity material. We thus begin by deriving the travel time curves for such a model, which show when seismic waves arrive at a particular distance from a seismic source. The travel times, especially those of waves that are critically refracted at the interfaces, are used to find the velocities of the layers and underlying halfspace and the layer thicknesses. As a result, this technique is called *refraction seismology*.

Refraction seismology is used on vastly differing scales. Near-surface structure at depths less than 100 meters can be studied using a sledge hammer or a shotgun as a source and a single receiver. Similar methods are used to study the crust and the upper mantle, with earthquake or explosion sources and many receivers at distances of hundreds of kilometers.

The simplest situation, shown in Fig. 3.2-1, is a layer of thickness h_0 , with velocity v_0 , overlying a halfspace with a higher velocity, v_1 . We write the velocities as “ v ” to indicate that the analysis applies for either P or S waves. There are three basic ray paths from a source on the surface at the origin to a surface receiver at x . The travel times for these paths can be found using Snell’s law.

The first ray path corresponds to a *direct wave* that travels through the layer with travel time

$$T_D(x) = x/v_0. \quad (1)$$

This travel time curve (Fig. 3.2-2) is a linear function of distance, with slope $1/v_0$, that goes through the origin.

The second ray path is for a wave reflected from the interface. Because the angles of incidence and reflection are equal,

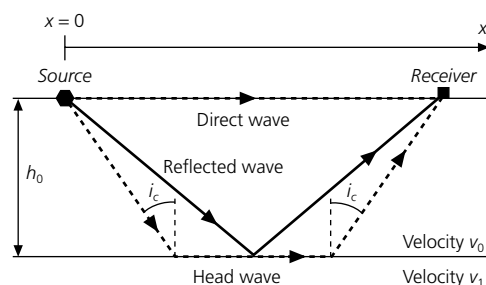


Fig. 3.2-1 Three basic ray paths for a layer over a halfspace model. The direct and reflected rays travel within the layer, whereas the head wave path also includes a segment just below the interface. For the head wave to exist, the layer velocity v_0 must be less than the halfspace velocity v_1 .

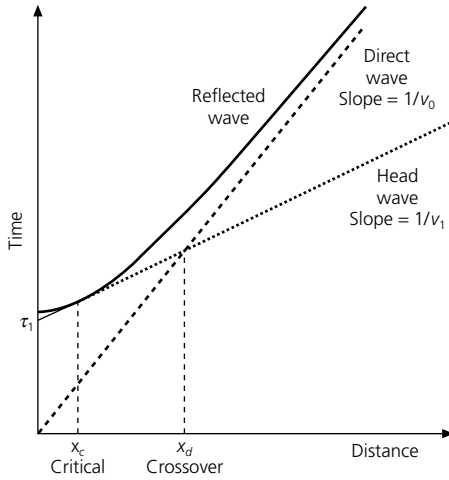


Fig. 3.2-2 Travel time versus source-to-receiver distance plot for the three ray paths in Fig. 3.2-1. The direct wave is the first arrival for receivers closer than the crossover distance x_d . Beyond x_d the head wave arrives first. The head wave exists only beyond the critical distance x_c .

the wave reflects halfway between the source and the receiver. The travel time curve can be found by noting that $x/2$ and h_0 form two sides of a right triangle, so

$$T_R(x) = 2(x^2/4 + h_0^2)^{1/2}/v_0. \quad (2)$$

This curve is a hyperbola, because it can be written

$$T_R^2(x) = x^2/v_0^2 + 4h_0^2/v_0^2. \quad (3)$$

For $x = 0$ the reflected wave goes straight up and down, with a travel time of $T_R(0) = 2h_0/v_0$. At distances much greater than the layer thickness ($x \gg h$), the travel time for the reflected wave asymptotically approaches that of the direct wave.

The third type of wave is the *head wave*, often referred to as a refracted wave. This wave results when a downgoing wave impinges on the interface at an angle at or beyond the critical angle. Its travel time can be computed by assuming that the wave travels down to the interface such that it impinges at the critical angle, then travels just below the interface with the velocity of the lower medium, and finally leaves the interface at the critical angle and travels upward to the surface. Thus the travel time is the horizontal distance traveled in the halfspace divided by v_1 plus that along the upgoing and downgoing legs divided by v_0 :

$$\begin{aligned} T_H(x) &= \frac{x - 2h_0 \tan i_c}{v_1} + \frac{2h_0}{v_0 \cos i_c} \\ &= \frac{x}{v_1} + 2h_0 \left(\frac{1}{v_0 \cos i_c} - \frac{\tan i_c}{v_1} \right). \end{aligned} \quad (4)$$

The last step used the fact that the critical angle (Section 2.5.5) satisfies

$$\sin i_c = v_0/v_1. \quad (5)$$

To simplify Eqn 4, we use trigonometric identities showing that

$$\cos i_c = (1 - \sin^2 i_c)^{1/2} = (1 - v_0^2/v_1^2)^{1/2} \quad (6)$$

and

$$\tan i_c = \frac{\sin i_c}{\cos i_c} = \frac{v_0/v_1}{(1 - v_0^2/v_1^2)^{1/2}}, \quad (7)$$

so Eqn 4 can be written

$$T_H(x) = x/v_1 + 2h_0(1/v_0^2 - 1/v_1^2)^{1/2} = x/v_1 + \tau_1. \quad (8)$$

Thus the head wave's travel time curve is a line with a slope of $1/v_1$ and a time axis intercept of

$$\tau_1 = 2h_0(1/v_0^2 - 1/v_1^2)^{1/2}. \quad (9)$$

This intercept is found by projecting the travel time curve back to $x = 0$, although the head wave appears only beyond the *critical distance*, $x_c = 2h_0 \tan i_c$, where critical incidence first occurs.

Because $1/v_0 > 1/v_1$, the direct wave's travel time curve has a higher slope but starts at the origin, whereas the head wave has a lower slope but a nonzero intercept. At the critical distance the direct wave arrives before the head wave. At some point, however, the travel time curves cross, and beyond this point the head wave is the first arrival even though it traveled a longer path. The *crossover distance* where this occurs, x_d , is found by setting $T_D(x) = T_H(x)$, which yields

$$x_d = 2h_0 \left(\frac{v_1 + v_0}{v_1 - v_0} \right)^{1/2}. \quad (10)$$

Hence the crossover distance depends on the velocities of the layer and the halfspace and the thickness of the layer.¹

Thus we can solve the inverse problem of finding the velocity structure at depth from the variation of the travel times observed at the surface as a function of source–receiver distance. This simple structure is described by three parameters. The two velocities, v_0 and v_1 , are found from the slope of the two travel time curves. We then identify the crossover distance and use Eqn 10 to find the third parameter, the layer thickness, h_0 . Alternatively, the layer thickness can be found from the reflection time or the head wave intercept (Eqn 9) at zero distance. Each of these methods exploits the fact that there is more than one ray path between the source and the receiver.

¹ A simple analogy is driving to a distant point by a route combining streets and a highway. If the destination is far enough away, it is quicker to take a longer route including the faster highway than a direct route on slower streets. The point at which this occurs depends on the relative speeds and the additional distance required to use the highway.

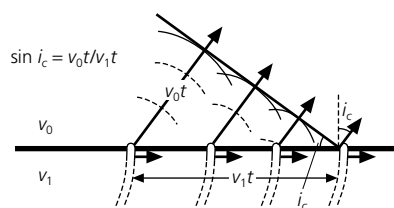


Fig. 3.2-3 Generation of an upgoing head wave by Huygens' sources due to a refracted pulse propagating along a boundary. The head wave travels in the upper layer at a slower velocity (v_0) than the refracted wave creating it, which travels in the layer below at velocity v_1 . (After Griffiths and King, 1981.)

Despite this solution's elegance, the basic assumption about the travel time of the head wave may seem unsatisfying, because it is unclear why energy should follow this path. However, the result conforms with observations — the experiment diagrammed in Fig. 3.2-1 yields an arrival whose travel time is given by Eqn 8. To understand why, we can view the head wave in several ways. As shown in this chapter's problems, it corresponds to a minimum time path between the source and the receiver, so, by Fermat's principle (Section 2.5.9), we expect such a wave. Another approach, using Huygens' principle (Section 2.5.10), is to consider the refracted wave traveling horizontally below the boundary at the velocity of the half-space, generating spherical waves that propagate upward in the lower-velocity layer (Fig. 3.2-3). The spherical waves interfere to produce upgoing plane waves that leave the interface at the critical angle.² However, our analysis of postcritical incidence (Section 2.6.4), which showed that an evanescent wave propagates along the interface, does not fully describe the head wave. A more sophisticated analysis than is appropriate here shows that the geometry in Fig. 3.2-1 gives the head wave's travel time, but not its amplitude, because geometrical optics are not applicable. Thus, although the energy propagation is more complicated than along the geometric ray path, the travel time predicted is correct.

Seismic refraction data led A. Mohorovičić³ in 1909 to one of the most important discoveries about earth structure. Observing two P arrivals (Fig. 3.2-4), he identified the first as having traveled in a deep high-velocity (7.7 km/s) layer, and the second as a direct wave in a slower (5.6 km/s) shallow layer about 50 km thick. These layers, now identified around the world, are known as the *crust* and the *mantle*. The boundary between them is known as the Mohorovičić discontinuity, or Moho. We now denote the head wave as P_n and the direct wave as P_g ("g" for "granitic"). Corresponding arrivals are also observed for S waves. The Moho, which defines the boundary

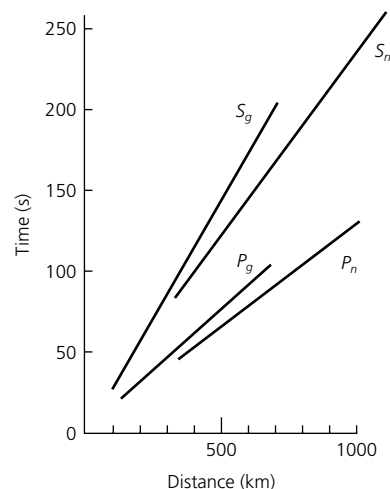


Fig. 3.2-4 Schematic of Mohorovičić's results showing the existence of a distinct crust and mantle. The travel time curves are labeled using modern nomenclature: the direct waves are P_g and S_g , and the head waves are P_n and S_n . (After Bonini and Bonini, 1979. *Eos*, 60, 699–701, copyright by the American Geophysical Union.)

between the crust and the mantle, has been observed around the world. One of the first steps in studying the nature of the crust is characterizing the depth to Moho, or crustal thickness, and the variation in P_n velocity from site to site.

Travel time plots for refraction experiments can be made by displaying seismograms in *record sections*. Because seismograms are functions of time, aligning several as a function of distance yields a travel time plot showing the different arrivals. Figure 3.2-5 shows a record section of a profile of seismograms recorded in England from explosive sources. In addition to P_n and P_g , the reflection off the Moho, known as P_mP , is well recorded. As expected, the direct and head wave travel times are linear with distance, whereas the reflection has a hyperbolic curvature. The figure is plotted as a *reduced travel time plot*, in which the time shown is the true time minus the distance divided by a constant velocity. This reduces the size of the plot, and makes waves arriving at the reducing velocity appear as a line parallel to the distance axis.

The geometry discussed here can correspond to different physical experiments. A single source can be recorded simultaneously at receivers at different distances. Alternatively, multiple sources at different distances can be recorded by a single receiver at different times. A single receiver can be moved away from a fixed source, so the same source is recorded at different distances. Similarly, a source can be moved away from a fixed receiver. Results of various experiments can be combined, using the principle of *reciprocity*, which states that the travel time is unchanged if the source and the receiver are interchanged. As a result, we can use travel time measurements without considering whether the source was at one position and the receiver at another, or the reverse. Moreover, because earth structure presumably is not changing during the experiment, data collected at different times can be combined.

² This situation is analogous to a bow wave from a boat or a supersonic wave from a jet airplane, in that the energy source travels faster than the wave it produces.

³ Andrija Mohorovičić (1857–1936), working in Zagreb, Croatia (then part of the Austro-Hungarian Empire), studied travel times from earthquakes in the region using recently invented pendulum seismographs.

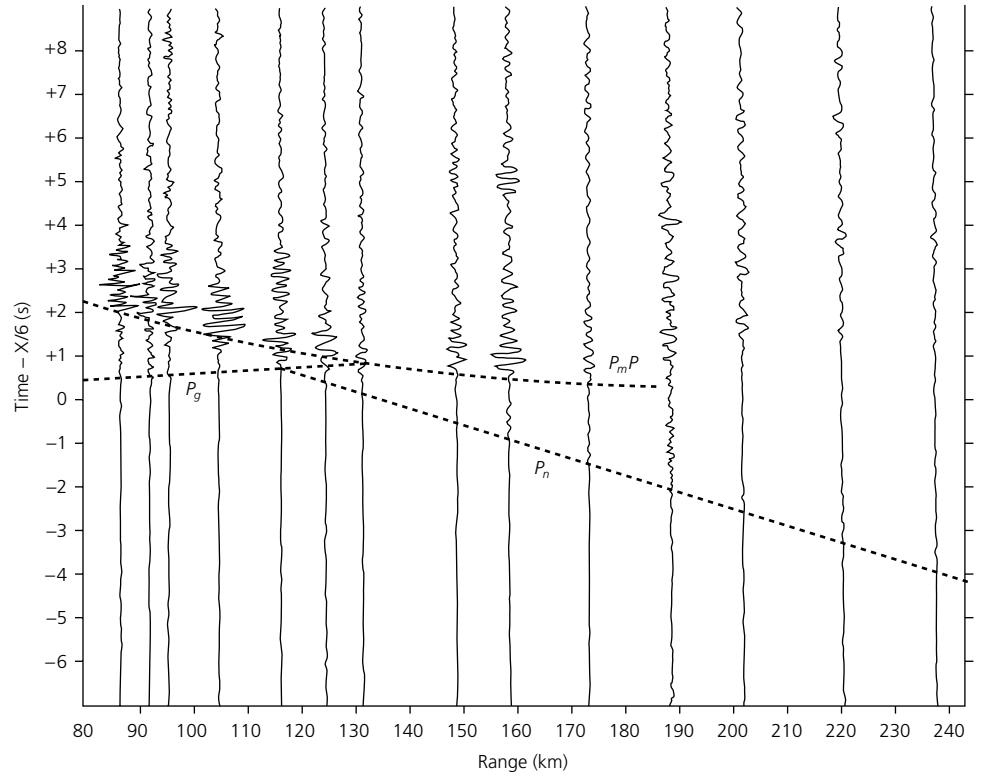


Fig. 3.2-5 Seismograms from a refraction profile, plotted with a reducing velocity of 6 km/s. The direct wave P_g , Moho head wave P_n , and Moho reflection P_mP are observed. P_g does not asymptotically approach P_mP as in Fig. 3.2-2 because the crust, instead of being homogeneous, has increasing velocity with depth. (Bott *et al.*, 1970. From *Mechanism of Igneous Intrusion*, ed. G. Newall and N. Rast, © 1970 by John Wiley & Sons Ltd. Reproduced by permission.)

Refraction data often show other arrivals in addition to P_g , P_n , and P_mP . Figure 3.2-6 shows a record section that also contains head waves P_i and P_n2 from boundaries within the crust and the mantle and P_iP , a reflection off a mid-crustal interface, which is analogous to the P_mP reflection off the Moho.

Such data require a model with multiple layers. Figure 3.2-7 shows a model in which a head wave arises at each interface where the velocity increases with depth. The travel time curve for a head wave at the top of the n^{th} layer is a line with slope $1/v_n$, that can be extrapolated to its intercept on the t axis, τ_n , and written

$$T_{H_n}(x) = x/v_n + \tau_n, \quad (11)$$

where, by analogy to the layer over the halfspace case (Eqn 9),

$$\tau_n = 2 \sum_{j=0}^{n-1} h_j (1/v_j^2 - 1/v_n^2)^{1/2}. \quad (12)$$

The thickness of successive layers can be found by starting with the top layer, whose thickness h_0 is given by Eqn 9 or 10, and continuing downward using the iterative formula

$$h_{n-1} = \frac{\tau_n - 2 \sum_{j=0}^{n-2} h_j (1/v_j^2 - 1/v_n^2)^{1/2}}{2(1/v_{n-1}^2 - 1/v_n^2)^{1/2}}. \quad (13)$$

Thus for two layers over a halfspace, the thickness of the second layer is found by setting $n = 2$, so

$$h_1 = \frac{\tau_2 - 2h_0(1/v_0^2 - 1/v_2^2)^{1/2}}{2(1/v_1^2 - 1/v_2^2)^{1/2}}. \quad (14)$$

A few examples illustrate some other complexities of refraction experiments. If the velocity increases with depth, the travel time curve for the head wave at the top of each successive layer has a shallower slope. By contrast, a low-velocity layer (Fig. 3.2-8) does not cause a head wave, so the travel time curve does not have a first arrival with the corresponding velocity, and depths to interfaces calculated using Eqn 13 are incorrect. Another possible problem occurs if a layer is thin or has a small velocity contrast with the one below it. Although a head wave results, it may never appear as a first arrival (Fig. 3.2-9), causing a *blind zone* that can be missed in the interpretation.

3.2.2 Dipping layer method

The refraction method can also be applied if the interfaces between layers are not horizontal. Conducting a *reversed profile* yields the travel times for ray paths in both the down-dip and the up-dip directions. This can be done using receivers on either side of a source, sources on either side of a receiver, or both. In this geometry, the depths to the interface below the source and the receiver differ due to the dip angle, θ . Consider

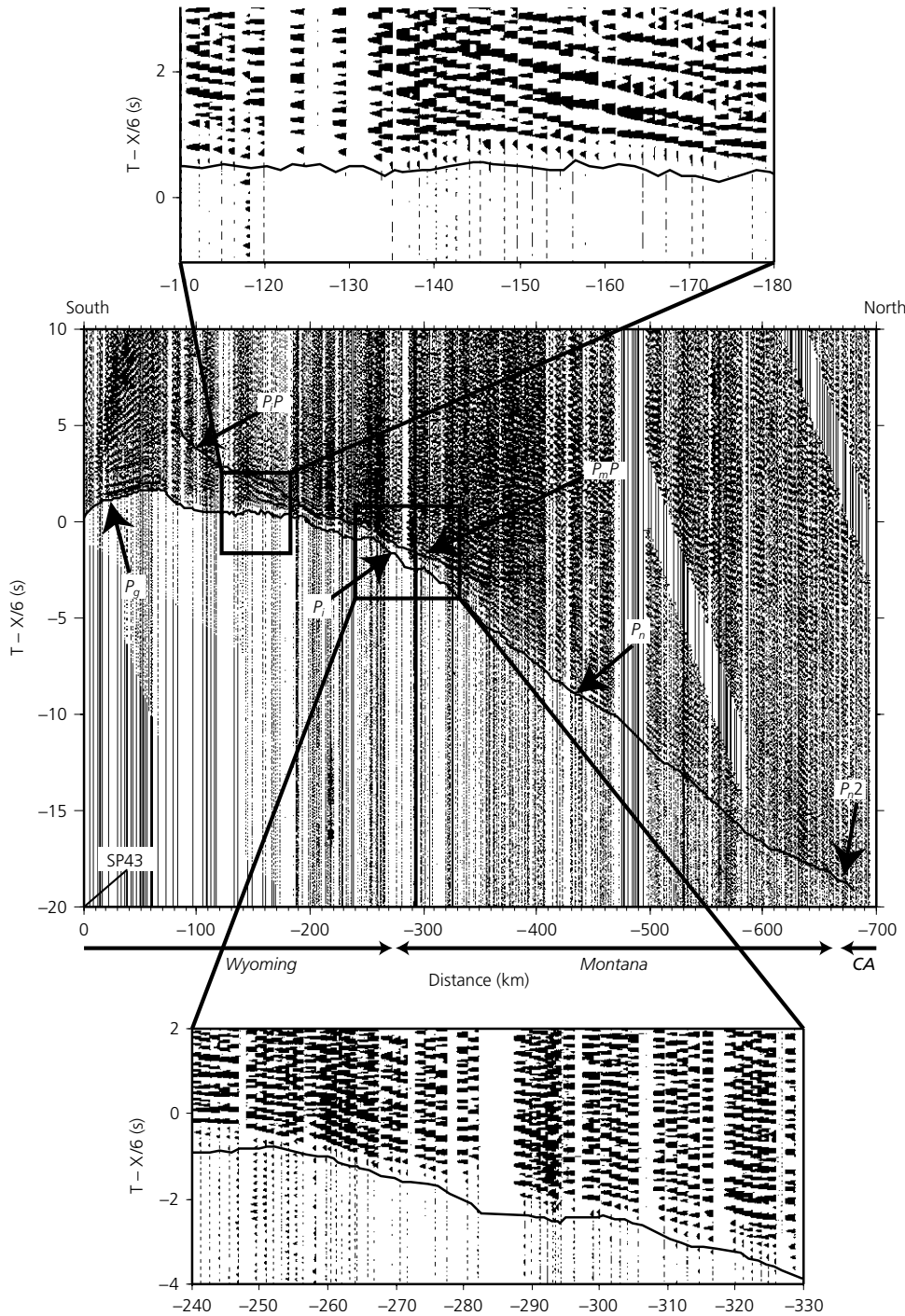


Fig. 3.2-6 Seismic refraction record section, plotted with a reducing velocity of 6 km/s. In addition to P_g , P_n , and P_mP , there are also arrivals P_i and P_n2 interpreted as head waves from boundaries within the crust and the mantle, and P_iP , interpreted as a reflection off a mid-crustal interface. (Snelson *et al.*, 1998.)

the down-dip ray path (Fig. 3.2-10) from a source, below which the perpendicular distance to the interface is h_d , to a receiver at a distance x , below which the perpendicular distance to the interface is $(h_d + x \sin \theta)$. The travel time for the head wave in the down-dip direction is the sum of the distance

along the interface divided by v_1 plus that for the upgoing and downgoing legs divided by v_0

$$T_d(x) = \frac{x \cos \theta - (2h_d + x \sin \theta) \tan i_c}{v_1} + \frac{(2h_d + x \sin \theta)}{v_0 \cos i_c}. \quad (15)$$

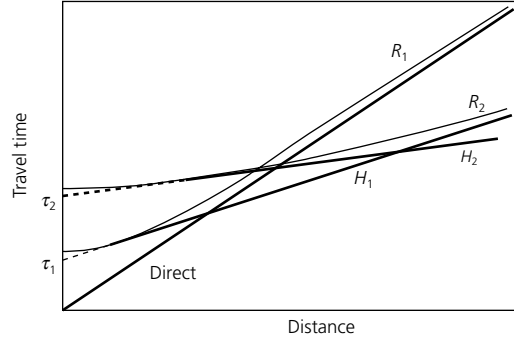


Fig. 3.2-7 Ray paths and travel times for a multilayered model in which velocity increases with depth. Each layer gives rise to a head wave H_i , whose intercept on the time axis is τ_i , and a reflection R_i . The direct wave arrival is also shown.

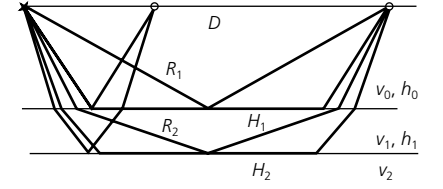


Fig. 3.2-8 Travel time curves, showing first arrivals only, for a model with three layers over a halfspace. Because the middle layer is a low-velocity layer with $v_1 < v_0$, no head wave arises at its top.

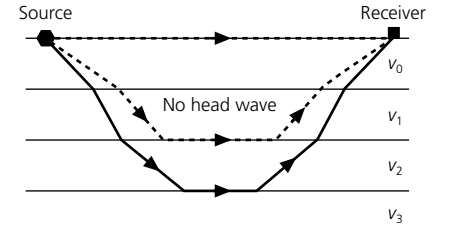
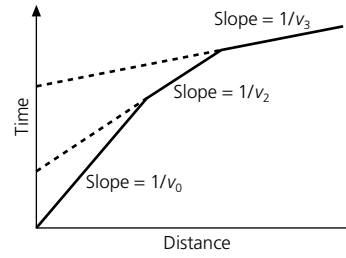


Fig. 3.2-9 Travel time curves, showing first arrivals only, for a blind zone geometry where the head wave from the top of layer 1 is never the first arrival because this layer is too thin.

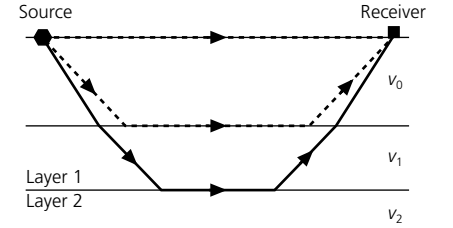
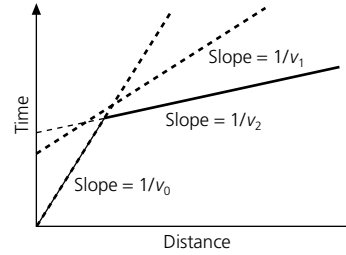
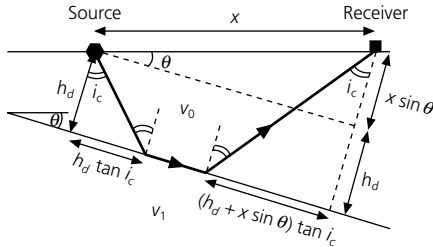


Fig. 3.2-10 Head wave ray path in the down-dip direction for a dipping interface over a higher-velocity halfspace. The layer thickness is measured perpendicular to the interface.



For the flat case, ($\theta = 0$), this is just Eqn 4. Simplifying using Eqns 5 and 7 yields

$$T_d(x) = \frac{x \cos \theta \sin i_c}{v_0} + \frac{(2h_d + x \sin \theta)(1 - \sin^2 i_c)}{v_0 \cos i_c} \\ = \frac{x \sin(i_c + \theta)}{v_0} + \frac{2h_d \cos i_c}{v_0} = \frac{x}{v_d} + \tau_d, \quad (16)$$

which is a straight line with slope $1/v_d$ and intercept τ_d .

Similarly, the travel time for the head wave in the up-dip direction is

$$T_u(x) = \frac{x \sin(i_c - \theta)}{v_0} + \frac{2h_u \cos i_c}{v_0} = \frac{x}{v_u} + \tau_u, \quad (17)$$

where $h_u = h_d + x \sin \theta$ is the perpendicular distance to the interface below the receiver (Fig. 3.2-10). Thus the apparent velocities, corresponding to the slopes of the head wave travel time curves, differ in the up-dip and down-dip directions by a factor depending on the dip angle,

$$v_u = v_0 / \sin(i_c - \theta) \quad v_d = v_0 / \sin(i_c + \theta). \quad (18)$$

The apparent velocity in the up-dip direction is greater than the halfspace velocity, and that in the down-dip direction is smaller. The time axis intercepts

$$\tau_u = 2h_u \cos i_c / v_0, \quad \tau_d = 2h_d \cos i_c / v_0, \quad (19)$$

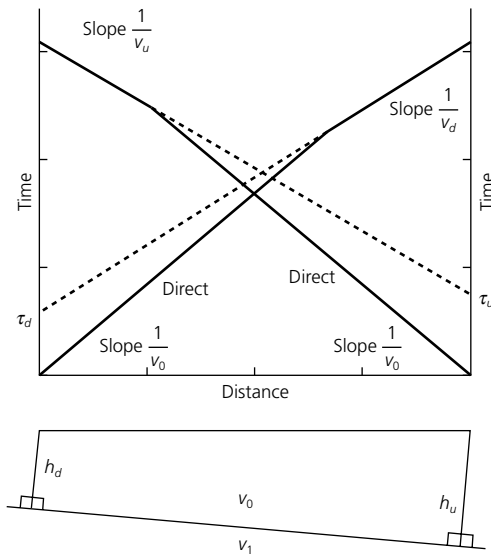


Fig. 3.2-11 Travel time plot for a reversed profile and its interpretation. The up-dip and down-dip slopes and intercepts differ.

also differ. The direct wave travel time is the same in both directions, so the crossover distances differ.

The results of a reversed profile are often displayed in the form shown in Fig. 3.2-11. The time axis is common to both directions, but distance is measured from one end of the axis for the up-dip experiment and from the other for the down-dip. The slopes of the direct and head wave travel times yield the dip angle

$$\theta = \frac{1}{2} \left(\sin^{-1} \frac{v_0}{v_d} - \sin^{-1} \frac{v_0}{v_u} \right) \quad (20)$$

and the critical angle

$$i_c = \frac{1}{2} \left(\sin^{-1} \frac{v_0}{v_d} + \sin^{-1} \frac{v_0}{v_u} \right). \quad (21)$$

The halfspace velocity v_1 is found from the critical angle and v_0 , and the intercept times then yield the layer thickness.

Two additional points about reversed profiles are worth noting. First, the different up-dip and down-dip head wave travel time curves do not imply that for a given pair of locations, it makes a difference whether the source is up-dip and the receiver down-dip, or the reverse (Fig. 3.2-12). By reciprocity, the two experiments give the same travel time. Thus, for a ray path connecting two points, it does not matter whether the wave travels up-dip or down-dip. By contrast, for two receivers at the same distance from a source, one up-dip and one down-dip, the travel times differ because the ray paths encounter the dipping interface at different depths. Similarly, the travel times

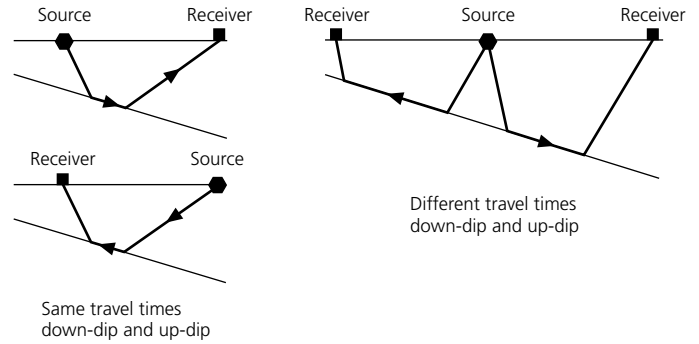


Fig. 3.2-12 *Left:* If the source and the receiver are interchanged on a reversed refraction profile, the travel time is unchanged. *Right:* Different up-dip and down-dip travel times occur because, for a given source position, waves going the same distance along the surface in opposite directions sample the dipping interface differently.

differ for two sources at the same distance from a receiver, one up-dip and one down-dip. If the dip were zero, then the travel times would be the same for all these cases because all ray paths encounter the interface at the same depth. Another way to view this is that for a flat geometry the travel time depends only on the distance between the source and the receiver. For a dipping geometry, the position as well as the separation matters, because the depth to the interface varies.

Second, the dip found from a reversed profile is not a true dip if the profile is not perpendicular to the strike of the layer. Instead, the measured dip is an apparent dip along the profile. The true dip can be found from the apparent dips along two reversed profiles that cross at a reasonably large angle, using a standard technique in structural geology.

3.2.3 Advanced analysis methods

Because the analysis above has been for simple geometries and uniform-velocity layers, refraction seismology might seem of little use in understanding the real earth. Fortunately, this is not the case. The simple geometries give models that fit data reasonably well and provide starting models for more sophisticated analyses.

Data from experiments showing travel times more complex than predicted by simple geometries can be interpreted with a computer program to trace rays using Snell's law through possible velocity structures. The predicted travel time curve is found by taking rays that arrive at a given distance, and integrating the slowness along their paths (Eqn 3.1.1). Figure 3.2-13 shows a record section and the inferred velocity structure for a refraction survey in central California. Ray paths calculated through the structure shown yield a good fit to the complicated travel time data. For example, the late arrivals about 8 km from the source are interpreted as resulting from a low-velocity region associated with a set of faults. The model

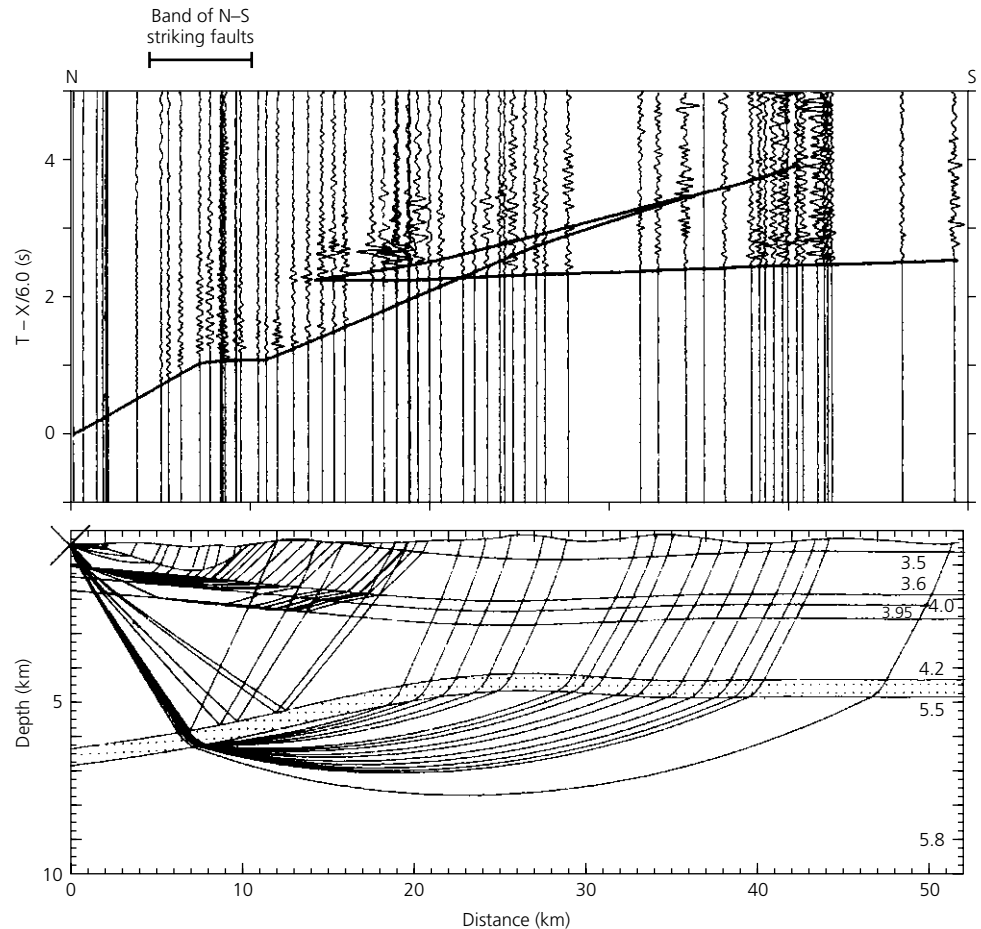


Fig. 3.2-13 Reduced travel time plot and ray tracing results for a seismic refraction survey. The solid line on the travel time plot shows the travel times predicted by the model. (Meltzer *et al.*, 1987. © Seismological Society of America. All rights reserved.)

also fits the travel times showing several velocity increases beyond this distance.

The restriction of uniform-velocity layers can also be surmounted. Geological instincts (a useful but occasionally unreliable tool) lead us to expect that rock types, and thus velocities, should often vary smoothly rather than in discrete jumps. Thus we expect velocity gradients with depth, rather than sharp interfaces. This possibility can be tested using advanced methods of analysis that predict both the travel times and the amplitudes of the expected arrivals. The amplitudes make it possible to distinguish gradients from uniform layers, even if the travel times predicted are the same. Although the methods are beyond our scope here, we discuss some results briefly.

To illustrate the relation between velocity structure and amplitudes, consider theoretical, or synthetic, seismogram record sections for the head wave, P_n , and Moho reflection, $P_m P$, predicted by two crustal models (Fig. 3.2-14). The seismograms were computed using a method known as reflectivity, which avoids the limitations of ray and plane wave analysis. The travel times are reduced at 8 km/s, and the direct wave is not shown. Both models have the same average velocity

structure, a 30 km-thick layer of 6.5 km/s material over an 8 km/s halfspace, so the travel times are similar. However, the amplitudes of the arrivals differ noticeably because the models have different fine structure near the Moho.

For the sharp Moho model (Fig. 3.2-14, *top*) the reflected wave is small for distances less than the critical distance (*subcritical* reflection), largest near the critical distance, and large for distances greater than critical (*supercritical*, *postcritical*, or *wide angle* reflections). Because the boundary is sharp, this amplitude behavior is similar to that predicted for plane waves (Fig. 2.6-11). $P_m P$ also shows the expected phase shift for reflection past critical incidence (Section 2.6.4). The head wave first appears near the critical distance, 83 km, and is small, as expected from the plane wave approximation that predicts no transmitted wave past the critical angle.

Figure 3.2-14 (*bottom*) shows the effect of velocity gradients above and below the Moho. Seismic energy trapped near the Moho yields larger P_n amplitudes than for the sharp Moho case. In addition, for subcritical distances, the reflection is smaller than without a gradient above the Moho, because it no longer reflects off a sharp interface. Hence the amplitudes

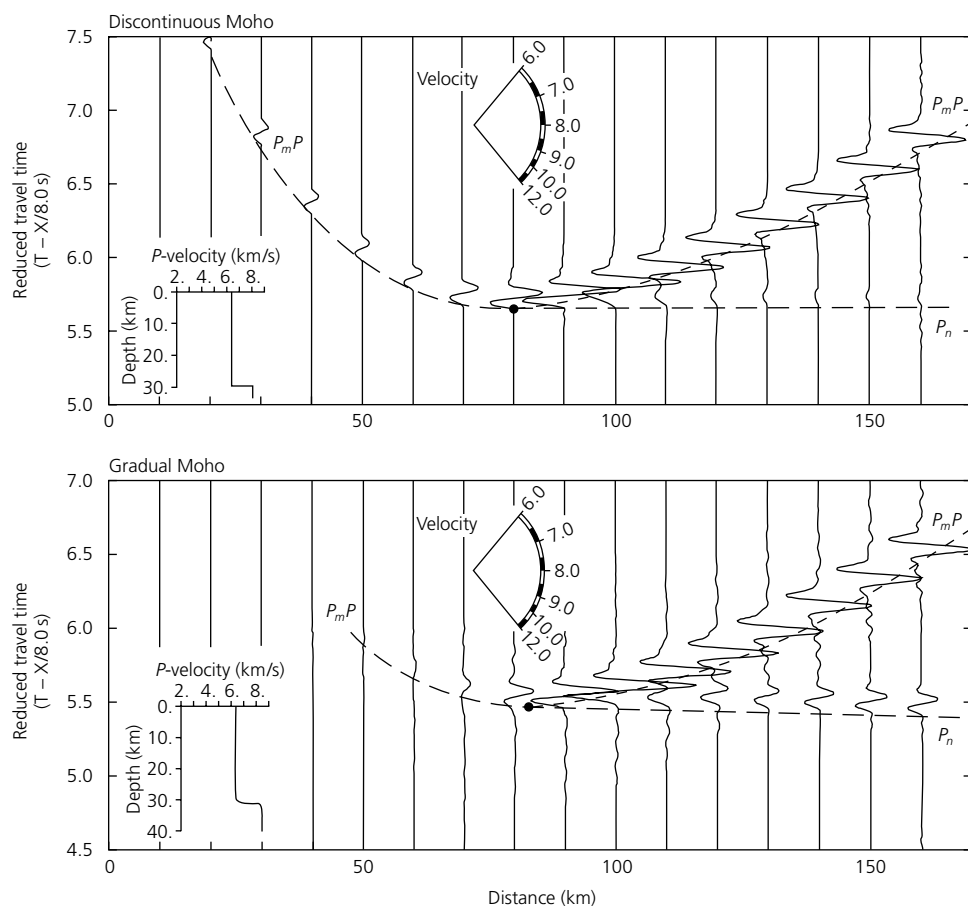


Fig. 3.2-14 Synthetic seismograms showing how the amplitudes of the head wave, P_n , and the reflected wave, P_mP , depend on the velocity structure at the Moho. Two cases with the same average-velocity structure are shown. At the *top* the Moho is a sharp transition, and at the *bottom* there are gradients above and below the Moho. The velocity scale shows the slopes of arrivals with different velocities. (After Braile and Smith, 1975.)

of P_n and P_mP indicate the presence or absence of gradients at the Moho.

Figure 3.2-15 illustrates these ideas for the oceanic crust and the mantle. Theoretical seismograms (Fig. 3.2-15, *center*) computed for a layered model that fits travel times predict strong reflections off the top of layer 3 (P_3P) and the Moho (P_mP). The observed data (Fig. 3.2-15, *bottom*) show strong P_mP reflections, suggesting a sharp Moho transition. However, strong P_3P reflections are not observed, implying that the transition between layers 2 and 3 is a gradient rather than a sharp jump. Thus, although the results of refraction studies are often reported as layered models that fit the travel times, amplitude studies are needed to show whether sharp interfaces exist.

An interesting point is that, because layers are distinguished from gradients by interpreting the amplitudes of seismic waves, this distinction depends on the wavelength of the wave used to study the structure. A reasonable approximation is that waves “see” only structures longer than their wavelengths. In other words, waves are affected by the medium properties averaged over their wavelengths. For example, the velocity structures in Fig. 3.2-16 appear identical to waves with a wavelength of 1 km, but look quite different for a wavelength of 1 m. Thus profile 3 appears as a sharp interface for waves with

wavelength 1 km, a gradient for 100 m wavelength, and a stack of layers for 10 m wavelength. The velocity structure depends on the wavelengths under discussion, so a velocity “gradient” is a structure that cannot be distinguished, with the wavelengths used, from one in which velocity changes smoothly. Similarly, an “interface” is a region that cannot be distinguished from a sharp velocity change with the wavelengths used.

3.2.4 Crustal structure

Information about crust and upper mantle structure around the world has been acquired by refraction surveys conducted on different scales. The size of the sources and the source-to-receiver distances increase with the depth of the structures being studied. Earthquakes or large explosions, including nuclear weapons tests, have enough energy to reveal the Moho. For example, the profile in Fig. 3.2-5, which showed clear Moho arrivals, was almost 250 km long and used sources containing 136 kg of explosive. Shorter profiles are used to study structure within the crust, as in Fig. 3.2-13. The recording stations are either permanent seismic stations or, in most cases, portable seismometers. Refraction studies are also conducted

Fig. 3.2-15 *Top*: Oceanic crust model with sharp transitions between layer 1 (water), layer 2 (unconsolidated sediment), layer 3 (crustal rock), and the mantle. *Center*: Synthetic seismograms for this model. P_2 , P_3 , and P_n are head waves from layers 2, 3, and the mantle. P_3P and P_mP are reflections off the tops of layer 3 and the mantle. *Bottom*: Data showing an absence of the large P_3P arrivals predicted by the layered model. (After Spudich and Orcutt, 1980. *Rev. Geophys. Space Phys.*, 18, 627–45, copyright by the American Geophysical Union.)

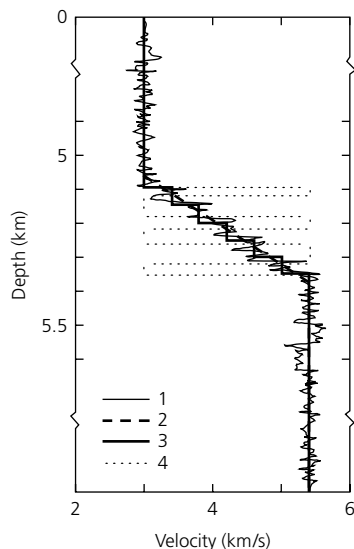
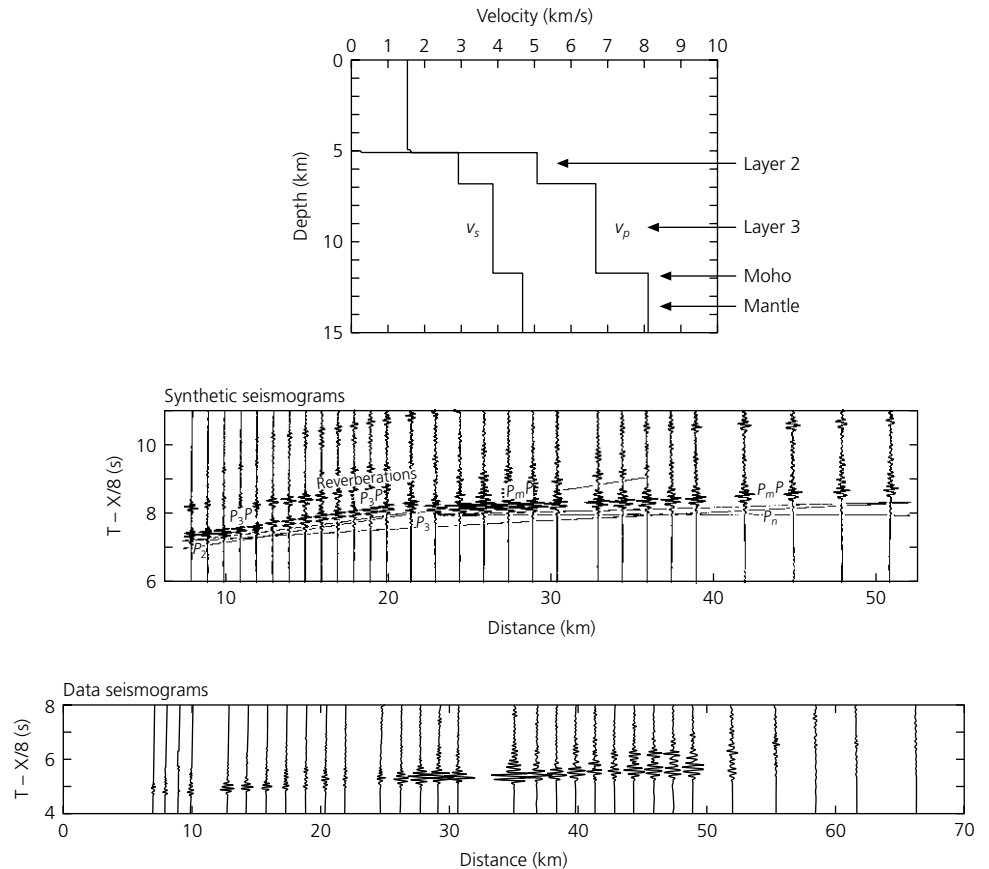


Fig. 3.2-16 Different velocity profiles that are indistinguishable when examined by using 1 km wavelength seismic waves, but distinguishable with much shorter wavelengths. (Spudich and Orcutt, 1980. *Rev. Geophys. Space Phys.*, 18, 627–45, copyright by the American Geophysical Union.)

at sea. In some cases, disposable sonobuoys or retrievable ocean bottom seismometers are deployed, and a ship steams away firing “shots.” In other cases, two ships are used. Marine refraction data (e.g., Fig. 3.2-15) are analyzed by treating the water as an upper layer of known velocity. The refraction results are combined with those from seismic reflection techniques, discussed in the next section, in which the velocity structure is derived from the travel times of subcritical reflections, rather than refractions. Refraction and reflection results are complementary and yield improved knowledge of structure.

The oceanic crust is about 7 km thick, and is relatively uniform from site to site, except at mid-ocean ridges. As a result, a single simple model like that in Fig. 3.2-15 is often applicable. By contrast, the continental crust is thicker and variable, as illustrated in Fig. 3.2-17 for a cross-section across the west coast of the United States. The thin crust beneath the Pacific Ocean thickens across the continent–ocean transition, such that beneath the coast ranges the Moho is about 25 km deep. Beneath the Sierra Nevada range, the depth to the Moho reaches 35–40 km. The refraction data also show complicated and variable-velocity structures within the crust. Thus the crust is not a uniform layer, or even a uniform set of layers, because

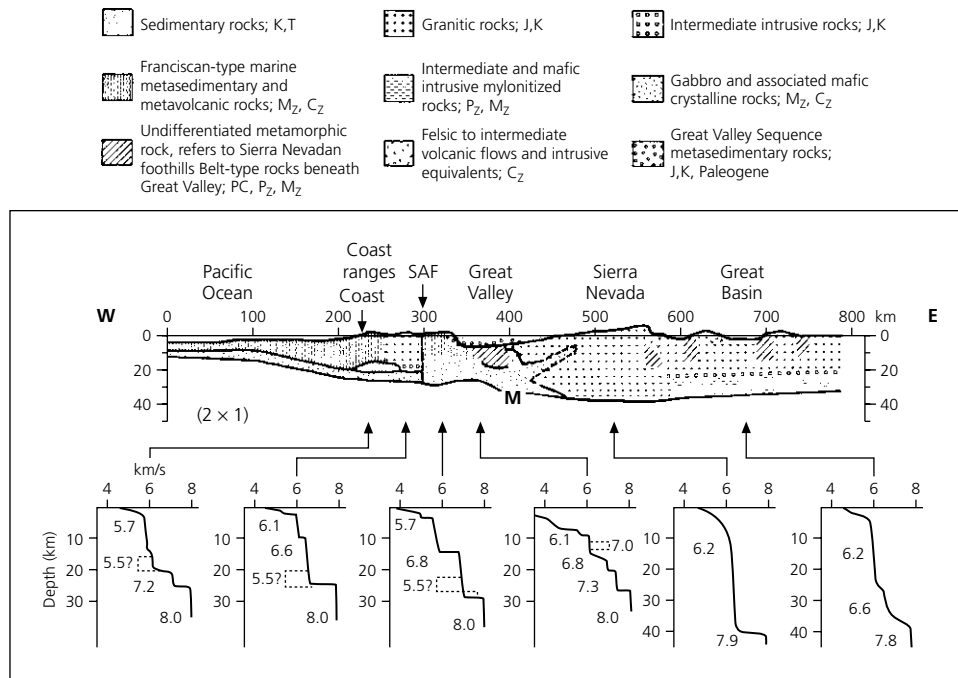


Fig. 3.2-17 Crustal velocity model and inferred geologic structure for a cross-section across the west coast of the USA. "SAF" denotes the San Andreas fault. Dashed lines indicate low-velocity zones. (After Mooney and Weaver, 1989. From *Geophysical Framework of the Continental United States*, ed. L. C. Pakiser and W. D. Mooney, with permission of the publisher, the Geological Society of America, Boulder, CO. © 1989 Geological Society of America.)

in some places it contains velocity gradients. Although early refraction studies suggested the existence of the Conrad discontinuity dividing the upper and lower crust, it now appears that high (greater than about 6.5 km/s)-velocity lower crust is present in some places but not in others. Furthermore, some areas show low-velocity zones within the crust.

Refraction studies show regional variations in crustal thickness and P_n velocities, as illustrated for North America in Fig. 3.2-18. East of $\sim 104^\circ\text{W}$, the crust is typically thick (~ 42 km), and P_n velocities are high (~ 8.1 km/s). To the west, the crust is often thinner, with lower P_n velocities. The thin crust and low P_n velocities beneath the Basin and Range province may reflect hotter material near the surface, consistent with active extension. As seen here and globally (Fig. 3.2-19), mountain ranges often have thick crust. The thick crust is thought to be due to isostasy, whereby the excess mass of the mountains is at least partially compensated by a crustal root with density less than that of the mantle.

The continental Moho can be modeled as a simple interface for the wavelengths used in most refraction studies. However, seismic reflection studies, with shorter wavelengths, sometimes show a laminated structure of high- and low-velocity layers (Fig. 3.2-20). In other cases, however, the Moho is not observed in reflection data. Some of these complexities may reflect difficulties associated with seismic reflection studies in laterally varying media (Section 3.3). Nonetheless, the Moho appears to be a complicated transition zone 0–5 km wide, with properties varying between locations (Fig. 3.2-21). Rather than regarding the Moho as the base of a homogeneous crustal

layer, it is better to view it as a zone where velocities increase rapidly with depth to values above about 7.7 km/s.

Velocity structures are often interpreted in terms of composition, as in Fig. 3.2-17. To do this, seismological results are combined with other geophysical data (e.g., gravity), geological fieldwork, and laboratory studies of the seismic velocities of rocks. The laboratory data show that velocity varies with composition, as shown in Fig. 3.2-22 for igneous rocks of the crust and upper mantle. Moreover, velocity increases with pressure and decreases with temperature. Inferences about composition are thus made by comparing predicted velocities to seismic observations. For pressures expected at greater depths, as for the lower mantle and core, laboratory experiments are more difficult, so thermodynamic calculations are also used to extrapolate experimental data to higher temperatures and pressures.

Such analyses imply that the upper continental crust has an average composition like granodiorite, whereas the upper oceanic crust is gabbroic.⁴ Historically, two types of models have been suggested for the Moho. In one, the Moho divides chemically different rocks, whereas in the other, it is a phase boundary separating rocks with the same bulk chemistry but different minerals. These models correspond to different combinations of rocks on either side. Two candidates for the lower continental crust are gabbro or rocks of intermediate composition in the granulite facies. The most popular candidate for the upper mantle is peridotite, which would make the Moho a

⁴ Some relevant rock and mineral nomenclature is summarized in Section 3.2.5.

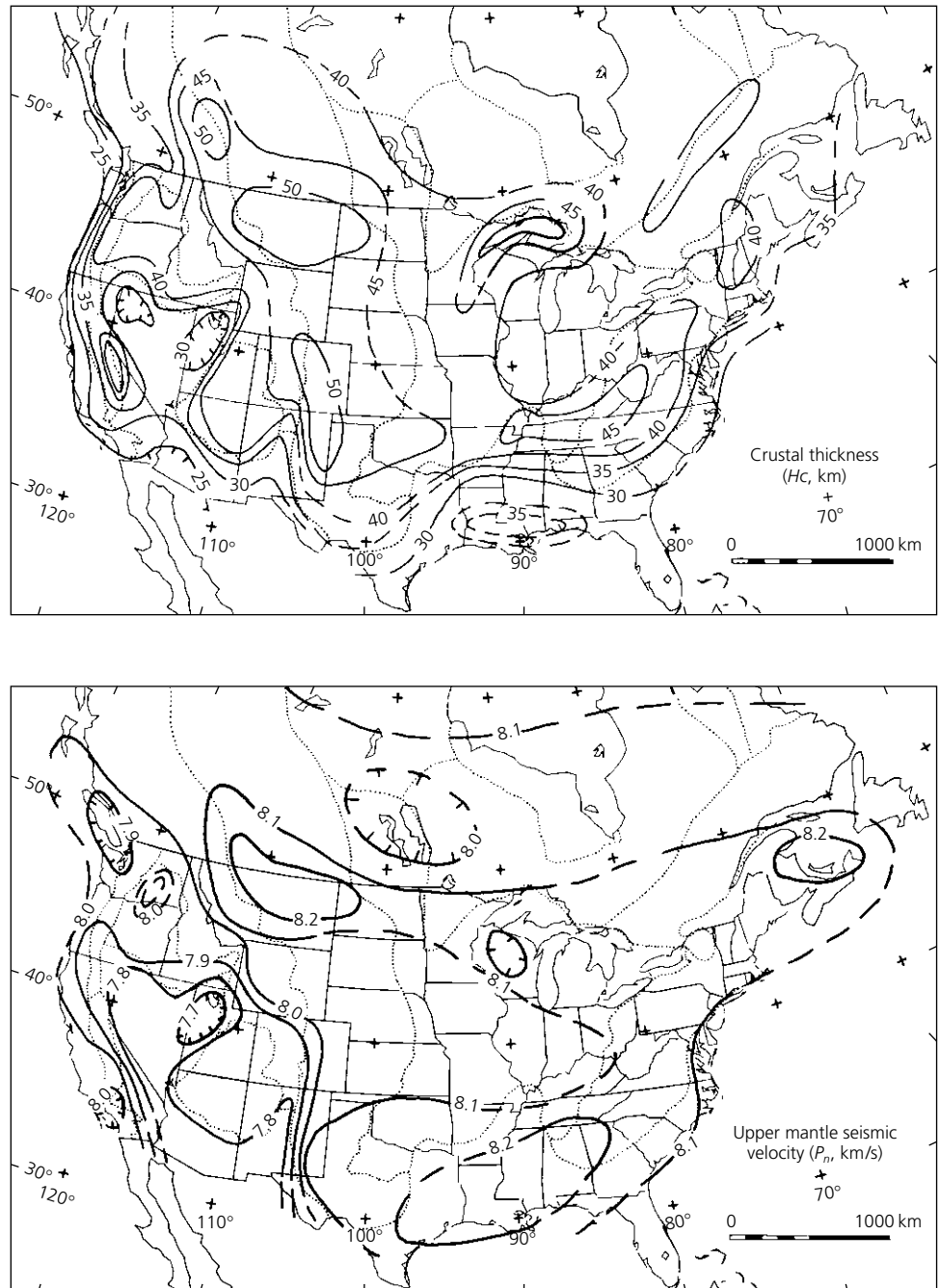


Fig. 3.2-18 Crustal thickness (depth to Moho) (*top*) and P_n velocity (*bottom*) maps for part of North America. Contour intervals are 5 km and 0.1 km/s. (Braile *et al.*, 1989. From *Geophysical Framework of the Continental United States*, ed. L. C. Pakiser and W. D. Mooney, with permission of the Geological Society of America, Boulder, CO. © 1989 Geological Society of America.)

compositional boundary. Another candidate is eclogite, a rock with the same bulk chemistry as gabbro, but denser mineral phases. If the upper mantle were eclogite and the lower continental crust gabbroic, the continental Moho would be a phase boundary. However, although eclogite and peridotite have similar seismic velocities, peridotite seems a more likely composition for the upper mantle. One of the reasons is that olivine, a major component of peridotite, yields anisotropic seismic velocities due to its crystal structure. Such anisotropic

P_n velocities are observed in the oceanic upper mantle and in some locations in the continental upper mantle (Section 3.6).

The status of the lower continental crust is more controversial. A granulite model is popular, but gabbro cannot be ruled out. Similarly, the origin of the laminated structure of the Moho is still unclear. Possible explanations include metamorphosed sediments, cumulate layering, tectonic banding, and lenses of partial melt. In any event, this structure seems to be laterally variable.

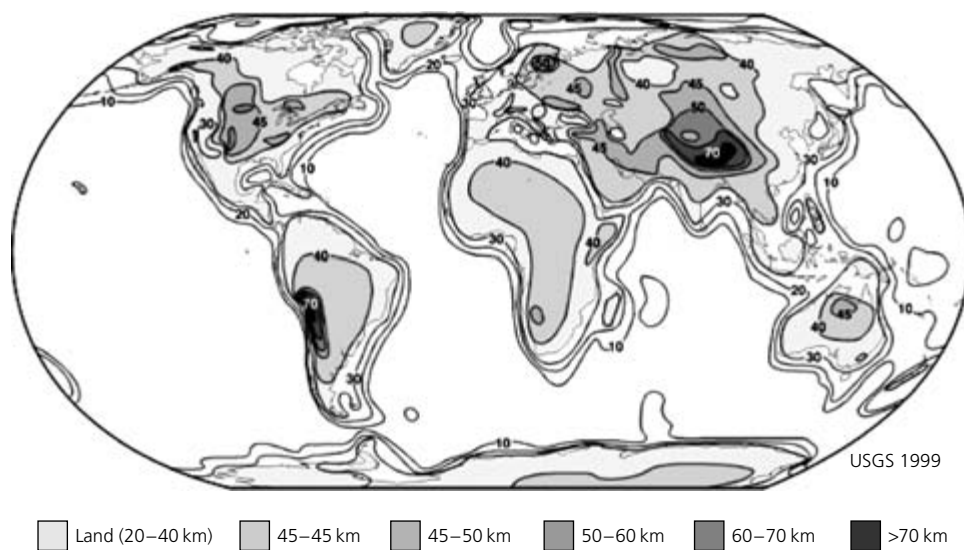


Fig. 3.2-19 Global map of crustal thickness. (Mooney *et al.*, 1998. *J. Geophys. Res.*, 103, 727–47. Copyright by the American Geophysical Union.)

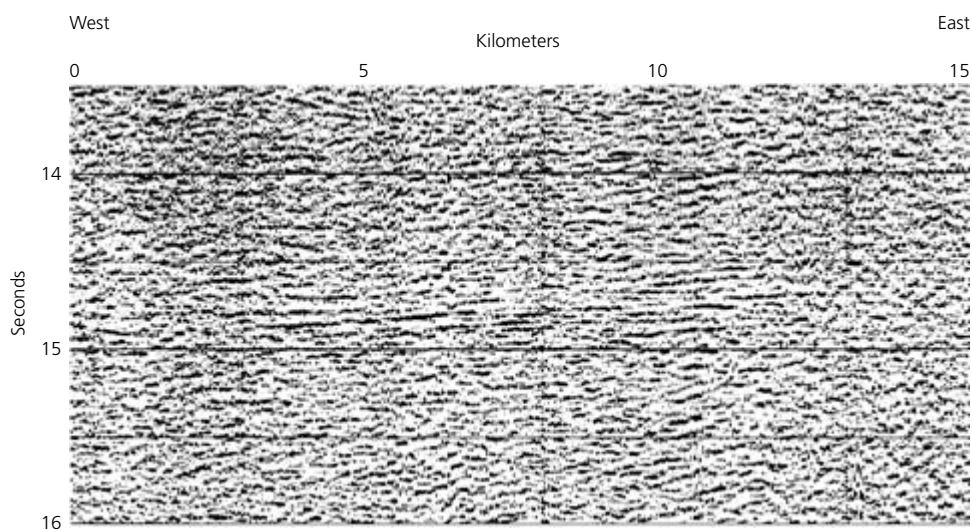


Fig. 3.2-20 Seismic reflection profile from the Wichita Mountains of southeastern Oklahoma. The “ringing” Moho reflections at 14.5–15 s in the middle of the section suggest that the Moho has a laminated velocity structure over several km. (Hale and Thompson, 1982. *J. Geophys. Res.*, 87, 4625–35, copyright by the American Geophysical Union.)

3.2.5 Rocks and minerals

Interpreting seismological results for the crust and mantle in terms of composition requires knowing something about rocks and the minerals that compose them. Although these are complicated subjects, we summarize a few essential terms.

For our discussions of crust and upper mantle structure, the most important rocks are the igneous rocks formed by cooling a molten magma. These rocks are classified primarily by the weight percent of silica, SiO_2 . A common nomenclature describes rocks as *acidic* or *silicic* for a weight percent of $\text{SiO}_2 > 66\%$, *intermediate* for 66–52%, *basic* or *mafic* for 52–45%, and *ultrabasic* or *ultramafic* for $< 45\%$.

Physical properties of rocks, such as density and seismic velocity, depend on their mineral composition. Figure 3.2-23

summarizes the major minerals in various rocks at near-surface temperatures and pressures. Because rock names refer to a range of compositions, those shown are averages. Rocks of the same composition have different names depending on whether they form at the earth’s surface (extrusive rocks) or below it (intrusive rocks). Hence an extrusive rock of gabbroic composition is a *basalt*.

Several important silicate (SiO_2 -bearing) minerals are mentioned in the figure. *Quartz* is pure SiO_2 . *Olivine* is a solid solution, $(\text{Mg}, \text{Fe})_2\text{SiO}_4$, whose composition varies from pure Fe_2SiO_4 (*fayalite*) to pure Mg_2SiO_4 (*forsterite*). Due to its crystal structure, olivine has anisotropic seismic velocities. *Pyroxene* is a solid solution with end members MgSiO_3 (*enstatite*), FeSiO_3 (*ferrosilite*), $\text{CaMg}(\text{SiO}_3)_2$ (*diopside*), and $\text{CaFe}(\text{SiO}_3)_2$ (*hedenbergite*), though only certain ranges of

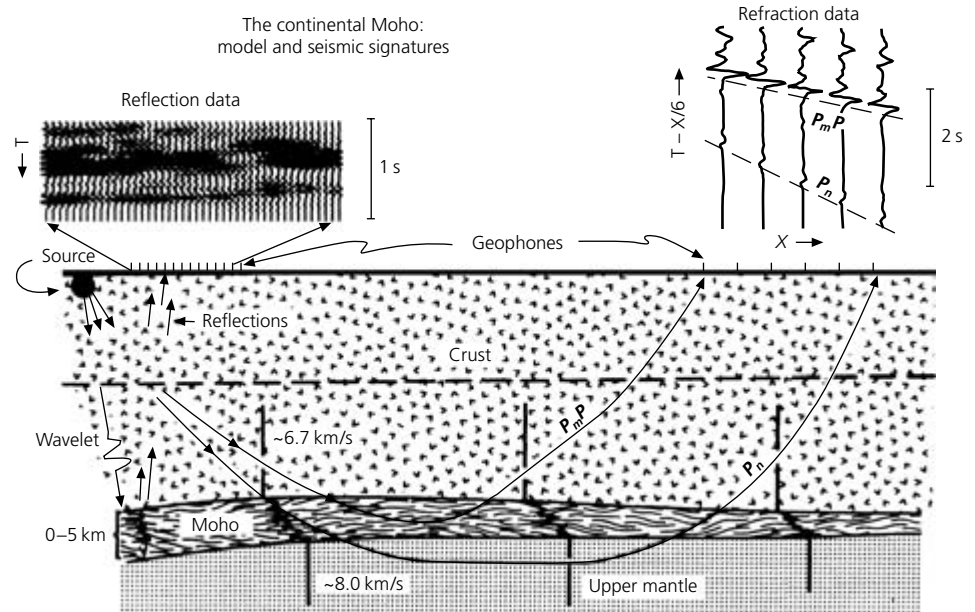


Fig. 3.2-21 Schematic model for the continental Moho as a laminated structure. Refraction studies using relatively longer wavelengths would show clear $P_m P$ and P_n arrivals, whereas reflection studies using shorter wavelengths would show reverberations. (Braile and Chiang, 1986. *Reflection Seismology*, 257–72, copyright by the American Geophysical Union.)

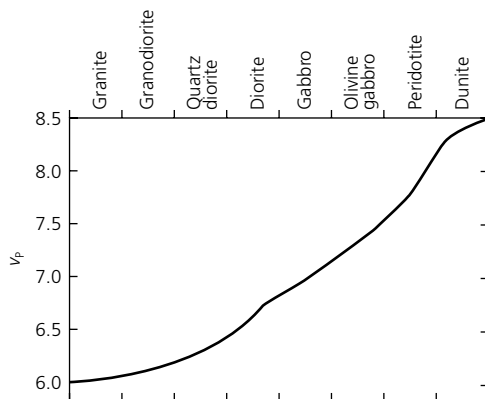


Fig. 3.2-22 Variation of P -wave velocity with lithology for crust and upper mantle rocks, at a pressure of 1.5 kbar (150 MPa). Velocity increases with decreasing silica content. (Fountain and Christensen, 1989. From *Geophysical Framework of the Continental United States*, ed. L. C. Pakiser and W. D. Mooney, with permission of the Geological Society of America, Boulder, CO. © 1989 Geological Society of America.)

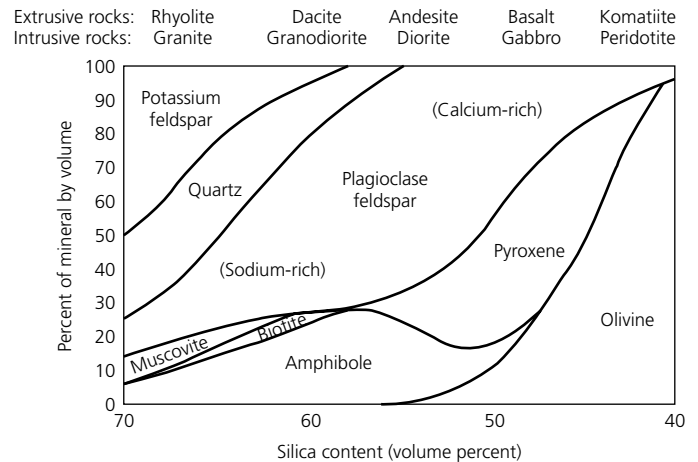


Fig. 3.2-23 Simplified igneous rock classification. Compositions are shown as the volume percent of major minerals for a rock of given silica content (horizontal axis). Thus a granodiorite of about 60% silica content contains about 20% amphibole, 5% biotite, 53% plagioclase feldspar, 17% quartz, and 5% potassium feldspar. Rock names are given for intrusive and extrusive forms.

compositions exist in nature. *Feldspar* is a solid solution with end members $\text{CaAl}_2\text{Si}_2\text{O}_8$ (*anorthite*), $\text{NaAlSi}_3\text{O}_8$ (*albite*), and KAlSi_3O_8 (*sanidine*, *orthoclase*, and *microcline*). The Na- and Ca-rich feldspars are called the *plagioclase feldspars*. A similar mineral group, the *amphiboles*, include *hornblende*, $\text{NaCa}_2(\text{Mg,Fe})_4(\text{Al,Fe})(\text{Si}_3\text{AlO}_{11})_2(\text{OH})_2$. *Biotite*, $\text{K}(\text{Mg,Fe})_3\text{Si}_3\text{AlO}_{10}(\text{OH})_2$, and *muscovite*, $\text{KAl}_2\text{Si}_3\text{AlO}_{10}(\text{OH})_2$, are in a group of minerals called *micas*. *Garnets* are minerals of the form $\text{A}_3\text{B}_2(\text{SiO}_4)_3$, where A is usually one of the ions Ca,

Mg, or Fe, and B is typically any of Al, Fe, or Cr. Garnets are comparatively dense, and thus significant for discussions of phase changes.

The figure describes rocks in terms of their mineralogy at surface conditions. With increasing pressure due to increasing depth in the earth, minerals transform to denser phases. Thus, for example, a gabbro containing plagioclase feldspar, pyroxene, and olivine transforms to a chemically identical eclogite rock containing quartz, pyroxene, and garnet. Hence

an argument against eclogite being a major component of the upper mantle is that, by contrast with peridotite, it does not contain olivine and would not yield the observed anisotropic P_n velocities. However, the gabbro-to-eclogite transformation may occur in subducting slabs (Section 5.4.2) and play a role in causing earthquakes there.

3.3 Reflection seismology

In the last section, we concentrated on the use of refracted arrivals to infer velocity structure with depth, and noted that reflected arrivals also contain valuable information for this purpose. Studies using the reflected arrivals, known as *reflection seismology*, determine velocities within the crust, and thus are essential in oil and gas exploration. As a result, data acquisition and processing methods have often been developed first by reflection seismologists. For example, digital data were generally used in exploration before they became common in earthquake studies. Similarly, because reflection data are densely sampled in space and time, and the mathematics of wave propagation in a layered medium is simpler than for a spherical earth, techniques are often first developed with reflection data. In this section we survey basic concepts in reflection seismology, some of which we later apply to earthquakes and the spherical earth.

3.3.1 Travel time curves for reflections

We first consider the simplest geometry: a flat layer of uniform-velocity material underlain by a halfspace with a higher velocity (Fig. 3.2-1). Although most applications use P waves, we write the velocity as “ v ” because the results also apply to S waves. For a layer of thickness h_0 with velocity v_0 , we saw in Section 3.2.2 that the travel time as a function of source-to-receiver distance, known as *offset* in reflection seismology, is

$$T(x)^2 = x^2/v_0^2 + 4h_0^2/v_0^2 = x^2/v_0^2 + t_0^2. \quad (1)$$

The travel time curve $T(x)$ is a hyperbola (Fig. 3.3-1) that intercepts the T axis at $t_0 = 2h_0/v_0$, the travel time at zero offset. This time is called the two-way vertical travel time, because the corresponding ray traveled vertically down to the reflector and back. Although this curve is the same as the “reflected wave” curve in Fig. 3.2-2, the convention in reflection seismology is to plot time increasing downward,¹ because later arrivals reflect deeper in the earth.

The layer velocity is found from the slope of the hyperbola. Because the slope decreases with increasing velocity, “flatter” travel time curves indicate higher velocities. To see this, note that a plot of $T(x)^2$ versus x^2 has slope $1/v_0^2$. Alternatively, the variation in travel time with offset is often stated in terms of

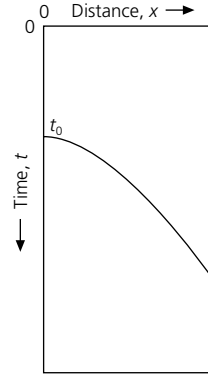


Fig. 3.3-1 The travel time curve for a reflection off a flat interface is a hyperbola, with the minimum at $x = 0$ corresponding to a vertical ray path. The slope is zero at $x = 0$ and increases with the offset distance.

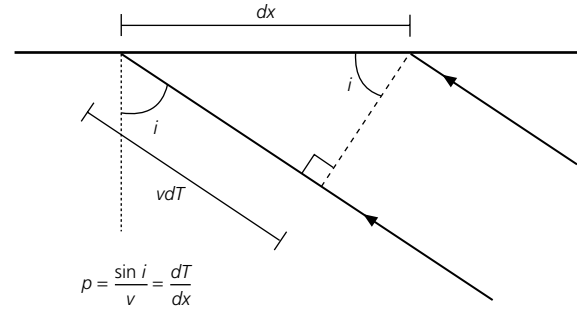


Fig. 3.3-2 Two rays showing the relationship between the angle of incidence, ray parameter, and the slope of the travel time curve for a flat medium.

normal moveout (NMO), the difference between the travel time at some offset and that at zero offset,

$$T(x) - t_0 = (x^2/v_0^2 + t_0^2)^{1/2} - t_0. \quad (2)$$

Once the velocity is found, the layer thickness is given by the vertical travel time.

To see the relation between the travel time curve and ray paths, consider the ray paths to two points dx apart, which differ in travel time by dT (Fig. 3.3-2). Because the ray paths differ in length by vdT , the angle of incidence can be found using

$$\sin i = \frac{vdT}{dx} \quad (3)$$

or, in terms of the ray parameter p (Section 2.5.7),

$$p = \frac{\sin i}{v} = \frac{dT}{dx}. \quad (4)$$

This is consistent with our earlier definition of the ray parameter as the reciprocal of the apparent velocity along the

¹ Earthquake seismologists generally follow the opposite convention.